

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30	U04		MODULAR DBB	K15AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN
40	U04		MODULAR DBB	K15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
50	U04	VAV, 817	MODULAR DBB	K15CC06	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
60	U04	VAW	MODULAR DBB	M15AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE R2BVR	0	0	BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.
70	U04	VAW	MODULAR DBB	M15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE R2BVR	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
80	U04	VAW	MODULAR DBB	M15EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	FF5	FLAT BOTTOM NIP OF FLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
90	U04	VAW	MODULAR DBB	M15EC02	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
100		VAW, 817	MODULAR DBB	M15ZC02	2	2	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350LF2 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
110	U04	VAW, 817	MODULAR DBB	M15ZC02	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350LF2 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
120	U04	VAW, 817	MODULAR DBB	K15MC06	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
130	U04	VAW, 817	MODULAR DBB	K15MC06L	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
140	U04	VAW, 817	MODULAR DBB	M15ZC02L	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF LTCS A350LF2 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
150	U04	VAW	MODULAR DBB	M15EC02L	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
160	U04	VAW	MODULAR DBB	M15EC02L	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
170		VB2					***** BALL VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
180											
180	U04		BALL	A15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0		
190	U04		BALL	A15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL RB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
200	U04	817	BALL	A15CC05	3	3	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0		
210	U04	817	BALL	A15CC06	3	3	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL RB LO	0	0		
220	U04	817	BALL	A15CC07-H1	4	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0		
230	U04	817	BALL	A15CC08	4	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL RB GO	0	0		
240	U04		BALL	B15AC01	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO	0	0		
250	U04		BALL	B15AC02	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING METAL FB LO	0	0		
260	U04	VA5	BALL	B15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB LO	0	0		
270	U04	VA5	BALL	B15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL FB LO	0	0		
280	U04	817	BALL	B15CC07-H1	3	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0		
290	U04	817	BALL	B15CC08	3	4	BALL 1500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL FB GO	0	0		
300							***** GLOBE VALVES *****				
300											
310	U04		GLOBE	L15AC48	0.75	1.5	GLOBE 1500# RF LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0		
320	U04		GLOBE	L15AC41	2	2	GLOBE 1500# RF LTCS A352 LCC INCONEL OS&Y HW	0	0		
330	U04	817	GLOBE	L15CC41	3	3	GLOBE 1500# CF LTCS A352 LCC INCONEL OS&Y HW	0	0		
340	U04	817	GLOBE	L15CC42-H1	4	4	GLOBE 1500# CF LTCS A352 LCC INCONEL OS&Y GO	0	0		
350							***** CHECK VALVES *****				
350											
360	U04	VC1	CHECK	C15AC50	0.75	1.5	CHECK PISTON 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM	0	0		
370	U04		CHECK	C15AC45	2	2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0		
380	U04	817	CHECK	C15CC52-H1	3	4	CHECK DUAL DBL FLG 1500# CF LTCS A352 LCC INCONEL TRIM	0	0		
390							***** PIPE *****				
390											
400											
400			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
410			PIPE	5358515	2	4	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
420			PIPE	5358513	6	6	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
430		812	PIPE	5358513	0.75	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
440			PIPE	5358528	12	12	PIPE WT 34.0MM SMLS LTCS A333 GR.6 BE	0	0		
450			PIPE	5358531	18	18	PIPE WT 47.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		
460							***** FITTINGS *****				
460											
470											
470			ELBOW	5392816	0.75	4	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELDED PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW: 1.5 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.3 MG 1.1 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 343°C	12/16/2024	BY OZAC	U04	Page 1 of 3	1500H1

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
480			ELBOW	5392822	0.75	4	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0	
490	U04	UK9	ELBOW	5392876	0.75	0.75	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0	
500	U04	UK9	ELBOW	5392870	0.75	0.75	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0	
510										
510			TEE	5392825	0.75	4	TEE BW SCH 160 LTCS A420 WPL6	0	0	
520			FLANGE	5392821	0.75	3	TEE RED BW SCH 160 LTCS A420 WPL6	1	4	
530	U04	UK9	TEE	5392879	0.75	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
540										
540			CAP	5392826	0.75	4	CAP BW SCH 160 LTCS A420 WPL6	0	0	
550	U04		CAP	5392880	6	6	CAP BW SCH XXS LTCS A420 WPL6	0	0	
560										
560			RED CONC	5392827	0.75	3	REDUCER CONC SCH 160 LTCS A420 WPL6	1	4	
570			RED ECC	5392828	0.75	3	REDUCER ECC SCH 160 LTCS A420 WPL6	1	4	
580			RED ECC	5392950	3	3	REDUCER ECC SCH XXS/SCH 160 LTCS A420 WPL6	6	6	
590			RED ECC	5392951	4	4	REDUCER ECC WT 24.0MM/SCH 160 LTCS A420 WPL6	8	8	
600			RED ECC	5392952	6	6	REDUCER ECC WT 34.0MM/SCH XXS LTCS A420 WPL6	12	12	
610			RED ECC	5392953	8	8	REDUCER ECC WT 34.0MM/24.0MM LTCS A420 WPL6	12	12	
620			RED ECC	5392954	12	12	REDUCER ECC WT 47.0MM/34.0MM LTCS A420 WPL6-W	18	18	
630			RED ECC	5392954	12	12	REDUCER ECC WT 47.0MM/34.0MM LTCS A420 WPL6-W	18	18	
640										
640							***** OLETS*****			
650										
650	U04		OLETS	5504159	0.75	1.5	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	2	4	
660	U04		OLETS	5504170	0.75	1.5	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	2	4	
670	U04		OLETS	5504298	1	1	NIPOFLANGE 1500# RF 34.0MM/SCH 160 LTCS A350 LF2 CL.1	12	12	
680	U04		OLETS	5504299	1	1	NIPOFLANGE 1500# RF 47.0MM/SCH 160 LTCS A350 LF2 CL.1	18	18	
690	U04	FF5	OLETS	5504321	0.75	1	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 160 LTCS A350 LF2 CL.1	0	0	
700										
700	U04		OLETS	5505392	0.75	1.5	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	4	
710	U04		OLETS	5504608	2	2	WELDOLET XXS/ SCH 160 LTCS A350 LF2 CL.1	6	6	
720										
720	U04	UJ2	ELBOFLG	5516080	1.5	1.5	ELBOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	4	4	
730										
730	U04	UD7	OLETS	5514772	0.75	1	THREDOLET 3000# LTCS A350 LF2 CL.1	2	4	
740										
740	U04		OLETS	5533071	0.75	1.5	SOCKOLET 6000# LTCS A350 LF2 CL.1	2	4	
750										
750							***** COUPLINGS, UNIONS & PLUGS*****			
760										
760	U04		COUPLING	5478807	0.75	1.5	COUPLING 6000# SW LTCS A350 LF2 CL.1	0	0	
770	U04		FLANGE	5478812	0.75	1	COUPLING RED 6000# SW LTCS A350 LF2 CL.1	1	1.5	
780										
780	U04		UNION	5478809	0.75	1.5	UNION 6000# SW LTCS STL A350 LF2 CL.1	0	0	
790										
790	U04	UD7	FITTING SCRD	5491303	0.75	1	PLUG ROUND HEAD SCRD LTCS A350 LF2 CL.1	0	0	
800										
800		816					***** FLANGES*****			
810										
810	U04	FA8	FLANGE	5599704	0.5	0.5	FLG SCRD (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
820										
820			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
830			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
840										
840	U04		FLANGE	5557397	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
850	U04		FLANGE	5557397	3	4	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
860	U04	UK9	FLANGE	5557505	0.75	0.75	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
870	U04		FLANGE	5557705	1.5	1.5	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH 160	0	0	
880	U04		BLIND	5615616	0.5	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
890	U04	817	FLANGE	5619838	3	4	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
900	U04	817	FLANGE	5619841	6	6	FLG COMPACT 1500# RJ LTCS A350 LF2 CL.1 SCH XXS	0	0	
910	U04	817	BLIND	5619785	3	6	FLG COMPACT BLIND 1500# RJ LTCS A350 LF2 CL.1 LINE CLASS 1500H1	0	0	
920										
920							*****ORIFICE FLANGES*****			
930										
930	U04		ORIFICE FLG	5624598	2	4	WN ORIF SET 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
940										
940							***** PIPE NIPPLES*****			
950										
950		812		0			NIPPLES BB TO BE CUT FROM PIPE			
960										
960							***** GASKETS*****			
970										
970			GASKET	5672251	0.5	4	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELDED PWHT FOR WT > 19MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW: 1.5 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 CODE: ASME B31.3 MG 1.1 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 343°C	12/16/2024	BY OZAC	U04	Page 2 of 3	1500H1

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980	U04	817	GASKET	5673480	3	6	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500H1	0	0	
990	U04	GA6	GASKET	5673021	2	3	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
1000	U04		GASKET	5674480	1.5	1.5	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
1010										
1010		B1					***** BOLTING *****			
1020										
1020		BA5	BOLTING	5675770	0.5	6	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1030										
1030							***** LINE BLANKS *****			
1040										
1040			SPEC BLIND	5749595	0.75	4	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
1050		817	SPEC BLIND	5750412	0.75	4	SPECTACLE BLIND COMPACT 1500# RJ LTCS A516 GR70	0	0	
1060										

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